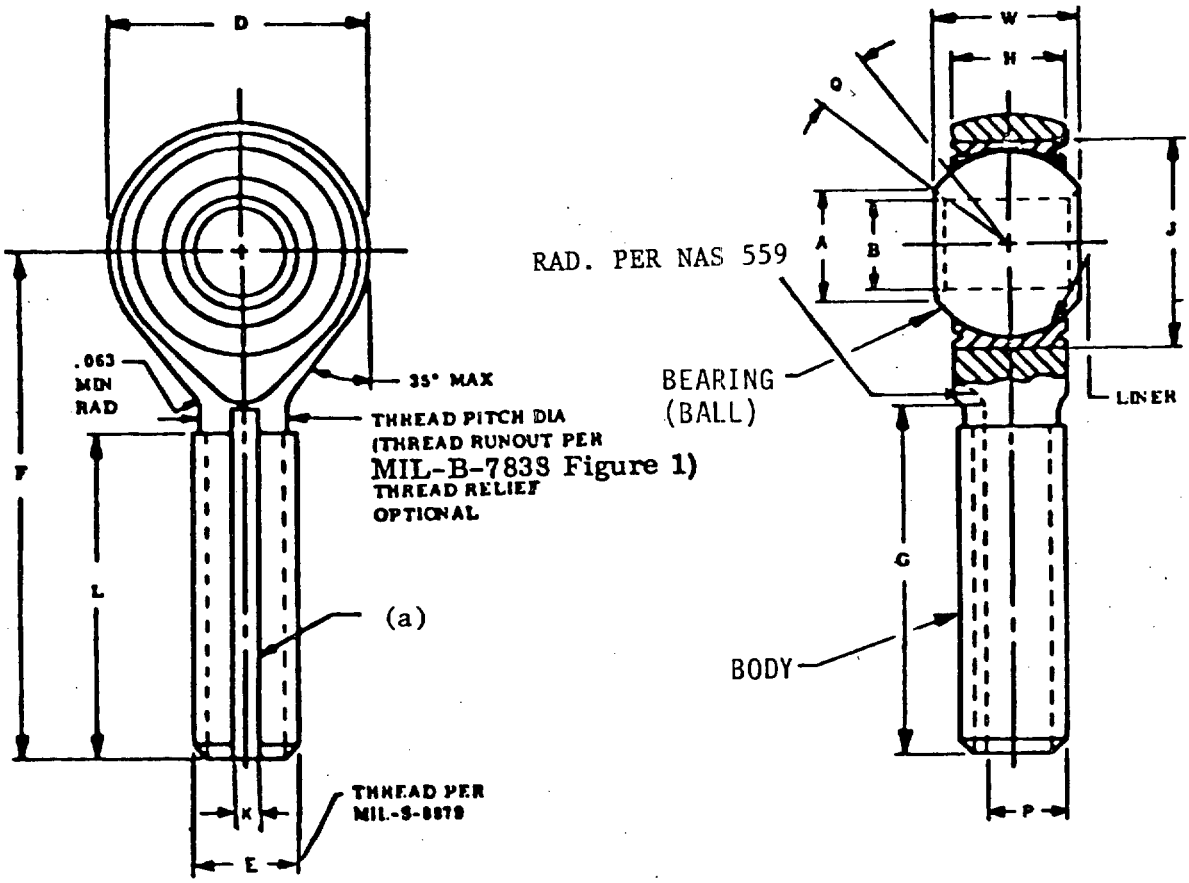
MIL-B-81935/1B
24 JUNE 1983
SUPERSEDING
MIL-B-81935/1A
31 July 1980

MILITARY SPECIFICATION SHEET

BEARING, PLAIN, ROD END, SELF-ALIGNING, SELF-LUBRICATING, WIDE, EXTERNALLY THREADED, -65°F TO +325°F

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the bearings described herein shall consist of this document and the latest issue of specification MIL-B-81935.



(a) KEYWAY SHALL BE ORIENTED ALONG THE CENTERLINE OF THE THREADED SHANK. THE PLANE OF THE BOTTOM OF THE KEYWAY SHALL BE PARALLEL TO THE PLANE MADE BY THE FACE OF THE ROD END HOOP.

FSC 3120

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TABLE I DIMENSIONS

DASH NUMBER	B BORE +.0000 -.0005	D OUTSIDE DIA ±.010	L THREAD LENGTH ±.031	E THREAD SIZE UNJF-3A	F CTRLINE BALL TO END ±.010	W BALL WIDTH +.000 -.002	H BODY WIDTH ±.005	A MIN	J MAX HOUSING I.D.	Q° MIN	G KEYWAY FLAT +.000 -.020	K KEYWAY(a) +.005 -.000	P KEYWAY(a) +.000 -.005
-3	.1900	.806	.968	.3125-24	1.562	.437	.337	.30	.6250	15	.980	.062	.268
-4	.2500												
-5	.3125	.900	1.187		1.875		.327	.36	.6875	14	1.270		
-6	.3750	1.025		.3750-24	1.938	.500	.416	.47	.8125	8	1.235	.093	.319
-7	.4375	1.150	1.281	.4375-20	2.125	.562	.452	.54	.9062	10	1.402		.383
-8	.5000	1.337	1.468	.5000-20	2.438	.625	.515	.61	1.0000	9	1.589		.445
-10	.6250	1.525	1.562	.6250-18	2.625	.750	.577	.75	1.1875	12	1.683	.125	.541
-12	.7500	1.775	1.687	.7500-16	2.875	.875	.640	.85	1.3750	13	1.808		.663
-14	.8750	2.025	2.000	.8750-14	3.375	.875	.765	1.00	1.6250	6	2.121	.156	.777
-16	1.0000	2.775	2.343	1.2500-12	4.125	1.375	1.015	1.27	2.1250	12	2.464	.187	1.136

(a) Keyway, when specified, is compatible with locking devices, MIL-B-81935/3 for sizes -3 thru -8, and NAS 513 for sizes -10 thru -16. Keyway tolerances not specified shall be in accordance with MIL-B-81935/3 or NAS 513 as applicable.

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TABLE II LOAD VALUES

DASH NO.	ULTIMATE STATIC LOAD (LBS)	FATIGUE (LOAD (LBS)	AXIAL PROOF LOAD (LBS)	WEIGHT MAX (LBS)	NO LOAD ROTATIONAL BREAKAWAY TORQUE IN-LBS	
					MIN	MAX
-3	2360	1470 (b)	1000	.072	.5	6
-4	4860	2380	1000			
-5	7180	2770 (c)	1100	.087	1	15
-6	8550	3670	1660	.136		
-7	12000	4800	1850	.183		
-8	19500	7680 (c)	2040	.278		
-10	21900	9180	2430	.424		
-12	29300	11600	2810	.639		
-14	34500	13100	3320	.963	1	24
-16	80300	30400	4340	2.546		

(b) BASED ON BOLT BENDING FATIGUE STRENGTH 180,000 PSI
(c) SHANK LIMITATION

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REQUIREMENTS:

MATERIAL:	
BODY:	ALLOY STEEL PER MIL-S-5000 (4340)
BEARING:	PER MIL-B-81820 AND MS14103
HARDNESS:	HEAT TREATMENT IN ACCORDANCE WITH MIL-H-6875
BODY:	Rc 39-42
SURFACE TEXTURE:	PER ANSI B46.1
BODY:	BORE: Ra 32 MAX SIDES OF THREAD AND ROOT AREA: Ra 32 THREAD RELIEF: Ra 63 ALL OTHER MACHINED SURFACES: Ra 125 MAX
PLATING:	CADMIUM PLATING PER QQ-P-416, TYPE II, CLASS 2
TOLERANCES:	UNLESS OTHERWISE SPECIFIED: DECIMALS ± .010; ANGLES ± 1/2°

DIMENSIONS IN INCHES, UNLESS OTHERWISE SPECIFIED.

BREAK SHARP EDGES AND CORNERS AND REMOVE ALL BURRS AND SLIVERS.

PART NUMBER: M81935/1 XX X X

- X — ADD L FOR LEFT HAND THREAD WHEN SPECIFIED
- X — KEYWAY (CODE K) WHEN SPECIFIED
- XX — BORE DIAMETER CODE IN MULTIPLES OF 1/16 INCH
- M81935/1 — BASIC PART NUMBER

Custodians:
Army - AV
Navy - AS
Air Force - 11

Review activities:
Army -
Air Force - 99
DLA - IS

Preparing activity:
NAVY - AS

(Project No. 3120-0610)

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